



IMO Selection 2025

First Exam – Solutions

Duration: 4.5 hours

Difficulty: The problems are ordered by difficulty.

Points: Each problem is worth 7 points.

Bern

May 10, 2025

Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

1. Let k be a positive integer. Leo owns an empty garden that measures 45 by 45 units. He wants to plant flowers in each of the 2025 unit squares, and can plant one square with flowers each morning. However, whenever flowers are planted in a square, then k days later in the evening, any orthogonally adjacent empty square will become infested with weeds. Infested squares can no longer have flowers planted in them.

Find the minimum value of k such that Leo can plant his entire garden with flowers.

Solution

The minimum value that still allows Frieder to plant his entire garden is $k = 45$.

Our strategy for $k = 45$ is to plant the flowers row by row, always starting with the leftmost square working our way over to the rightmost square. That is if we give the squares coordinates (i, j) with $1 \leq i, j \leq 45$. Then we plant flowers on square (i, j) on day $i + 45(j - 1)$. This works as for a given square (i, j) the neighbouring squares $(i, j - 1)$, $(i - 1, j)$, $(i + 1, j)$ and $(i, j + 1)$ will all be planted before or on day $i + 45j = (i + 45(j - 1)) + 45$.

Assume that Frieder has a strategy to plant all 2025 flowers. Consider the first day when Frieder has either already planted a flower in each row or already planted a flower in each column. Assume WLOG that something being planted in each row happens first.

Lemma No row is complete yet. That is, there exists no row that already has 45 flowers planted in it.

Proof. The last planted flower cannot be in that row (else one day earlier all rows would have been non-empty already). So a day earlier that row was already full and each column had at least one plant in it, a contradiction to our minimal choice. $\square$

So we have that each row contains at least one square with a flower already planted and at least one square which is still empty. So in each row there exists an empty square that neighbors a square with a flower already planted there. So, on this day, we have at least 45 squares with weeds growing in them, and it would take Frieder at least 45 days to plant a flower in each one of them. If k were strictly less than 45 this would be impossible.

Marking Scheme

- (A) **2P:** Find a construction to prove the upper bound of $k \leq 45$. (No points are awarded for simply stating the answer of $k = 45$.)
- (B) **1P:** State that the set of empty squares $\mathcal{B}$ bordering a flower has size at most k .
- (C) **3P:** Upper bound Version I
 - (C1) **2P:** Consider the first day all rows contain at least one flower or all columns contain at least one flower.
 - (C2) **1P:** Show that on that day there exists no full row or there exists no full column.
- (D) **3P:** Upper bound Version II
 - (D1) **2P:** Show that, given m flowers the shape that minimizes the border is connected and stairshaped.
 - (D2) **1P:** Consider the day after half the flowers have been planted. $\frac{n(n-1)}{2}$ or similar.

In the upper bound, the second point is only achievable if the first is achieved.

2. Let n be a positive integer and let $P(x)$ denote the polynomial

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$$

with real coefficients and $a_0 \neq 0$, whose n roots are all pairwise distinct, positive real numbers. Assuming that $P(x)$ divides $P(2x)P(x/2)$, prove that

$$\frac{a_{n-1}a_1}{a_0} \geq \frac{9n^2}{8}.$$

Solution

Let $P(x) = \prod_{k=1}^n (x - x_k)$, where $x_k \in \mathbb{R}_{>0}$ are the zeros of the polynomial. Using Vieta, we get that

$$a_{n-1} = -\sum_{k=1}^n x_k, \quad a_1 = \left(\sum_{k=1}^n \frac{1}{-x_k} \right) \prod_{k=1}^n (-x_k), \quad a_0 = \prod_{k=1}^n (-x_k).$$

Therefore, we have that

$$\frac{a_{n-1}a_1}{a_0} = \left(\sum_{k=1}^n x_k \right) \left(\sum_{k=1}^n \frac{1}{x_k} \right).$$

Let $z_1, \dots, z_r$ be the values appearing in $(x_k)_{k=1}^n$ ordered by value and $m_1, \dots, m_r$ be the multiplicity of root $z_1, \dots, z_r$. The condition that $P(x)$ divides $P(2x)P(x/2)$ translates to

$$\prod_{j=1}^r (x - z_j)^{m_j} \text{ divides } \prod_{j=1}^r (2x - z_j)^{m_j} \prod_{j=1}^r (x/2 - z_j)^{m_j} = \prod_{j=1}^r \left(x - \frac{z_j}{2}\right)^{m_j} (x - 2z_j)^{m_j}$$

Note that this implies that if z_j is a root of multiplicity m_j , then the sum multiplicity of $\frac{z_j}{2}, 2z_j$ is $\geq m_j$. (Lemma 1)

Let us construct the following graph: We construct the graph $G = (V, E)$ where $V = \{x_k | 1 \leq k \leq n\}$ and $\{x_i, x_j\} \in E(G) \iff x_j = 2x_i \vee x_i = 2x_j$. By Lemma 1, every vertex has a neighbour, thus every component has size at least 2. The vertices in a component of the graph thus correspond to a chain of the form $(x, 2x, \dots, 2^{l-1}x)$ (every vertex has degree ≤ 2 , and the graph has no cycles). Therefore, we can partition the roots $x_k, 1 \leq k \leq n$, into chains $X_j = (x_{1_j}, 2x_{1_j}, \dots, 2^{l_j-1}x_{1_j}), 1 \leq j \leq m$, such that $l_1 + \cdots + l_m = n$. Thus we can rewrite

$$\sum_{j=1}^n x_j = \sum_{i=1}^m \sum_{k=0}^{l_i-1} 2^k x_{1_i} = \sum_{i=1}^m (2^{l_i} - 1)x_{1_i} \text{ and } \sum_{j=1}^n \frac{1}{x_j} = \sum_{i=1}^m \sum_{k=0}^{l_i-1} \frac{1}{2^k x_{1_i}} = \sum_{i=1}^m \left(2 - \frac{1}{2^{l_i-1}}\right) \frac{1}{x_{1_i}}.$$

Note that by Cauchy-Schwarz:

$$\left(\sum_{k=1}^n x_k \right) \left(\sum_{k=1}^n \frac{1}{x_k} \right) \geq \left(\sum_{i=1}^m \frac{2^{l_i} - 1}{2^{(l_i-1)/2}} \right)^2$$

What remains to be shown is that $\frac{2^{l_i}-1}{2^{(l_i-1)/2}} \geq \frac{3}{2\sqrt{2}}l_i$. Equality holds for $l_i = 2$. Note that if we increase l_i by one, we can do following estimation

$$\frac{2^{l_i+1} - 1}{2^{l_i/2}} = 2^{(l_i+2)/2} - 2^{-l_i/2} > (\sqrt{2}-1)2^{(l_i+1)/2} + 2^{(l_i+1)/2} - 2^{-(l_i-1)/2} \geq (\sqrt{2}-1)2^{(l_i+1)/2} + \frac{3}{2\sqrt{2}}l_i.$$

For $l_i + 1 \geq 3$ we have: $(\sqrt{2} - 1)2^{(l_i+1)/2} \geq (\sqrt{2} - 1)2\sqrt{2} = 4 - 2\sqrt{2}$. A calculation shows:

$$4 - 2\sqrt{2} \geq \frac{3}{2\sqrt{2}} \iff 8\sqrt{2} \geq 11 \iff 128 \geq 121.$$

As $\sum_{i=1}^m l_i = n$, we have

$$\left(\sum_{i=1}^m \frac{2^{l_i} - 1}{2^{(l_i-1)/2}} \right)^2 \geq \left(\sum_{i=1}^m \frac{3}{2\sqrt{2}} l_i \right)^2 = \frac{9n^2}{8}.$$

Marking Scheme

For a full solution 7 points are awarded.

(A) **2P:** (starting observations)

(A1) **1P:** Reformulate $\frac{a_{n-1}a_1}{a_0}$ using the roots of the polynomial.

(A2) **1P:** Note that z_j being a root implies $2z_j$ or $z_j/2$ being a root.

(B) **3P:** (CS-bounding)

(B1) **1P:** Show that each root is part of such a chain of length at least 2.

(B2) **2P:** Apply Cauchy-Schwarz to the chains.

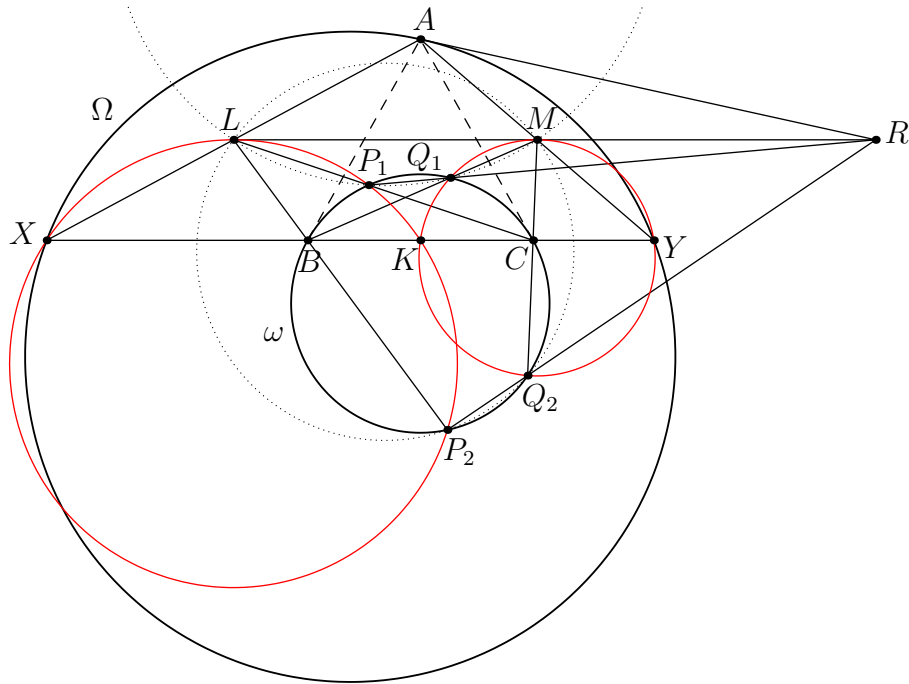
(C) **2P:** (finishing)

(C1) **1P:** Claim the expression is bounded from below by $\frac{3}{2\sqrt{2}}l_i$.

(C2) **1P:** Prove the above bounding.

3. Let Ω be a circle and ω a circle inside Ω . Point A lies on Ω , and tangents to ω through A touch ω at B and C . Let line BC intersect Ω at points X and Y . Let K, L and M be the midpoints of BC, AX and AY respectively. The circumcircle of XLK intersects ω at two points P_1 and P_2 , such that P_1 lies on the same side of BC as A . Similarly, the circumcircle of YMK intersects ω at two points Q_1 and Q_2 , such that Q_1 lies on the same side of BC as A . If lines P_1Q_1 and P_2Q_2 intersect at R , prove that RA is tangent to Ω .

Solution



Without loss of generality, we assume that X, B, K, C, Y lie on line BC in this order. The main claim is the following:

Claim: $L, B, P_2 / L, P_1, C / M, C, Q_2 / M, Q_1, B$ are collinear triples.

Proof. We show the claim for L, B, P_2 , the rest is symmetric. Let P be the second intersection of LB with ω . By the configuration condition, P doesn't lie on the same side of BC as A .

First, note that the midpoints of AB and AC have the same power with respect to ω and the degenerate circle (A) . Hence, the radical axis of ω and (A) is the A -midline of triangle ABC . But by Thales it is easy to see that L lies on this midline. Hence,

$$LX^2 = LA^2 = LB \cdot LP,$$

which by the tangent chord theorem means that $\angle LXB = \angle XPL$. But, as ABC is isosceles, $AK \perp BC$, meaning that AXK is a right triangle. As L is the midpoint of AX , $LX = LK$. Hence, $\angle LXX = \angle XKL$, and putting together with the above, $\angle XKL = \angle XPL$. In other words, $XLKP$ is cyclic, so $P = P_2$ as desired!

Note that $LM \parallel BC$, and therefore by Reim's theorem, we can check that LMQ_1P_1 and LMQ_2P_2 are cyclic quadrilaterals (reporting the parallelism to check that two angles subtended by the same side/diagonal are equal).

Determining now the radical center of (LMQ_1P_1) , (LMQ_2P_2) and ω , we get that P_1Q_1 , P_2Q_2 and LM are concurrent at R . (Notice that applying Pascal on $Q_1BP_2Q_2CP_1$ yields the same result). But in particular R lies on LM , the radical axis of A and ω , so

$$RA^2 = RP_1 \cdot RQ_1 = RL \cdot RM$$

(last equality due to the above cyclicity). This means that RA is tangent to (ALM) .

Lastly, there is a homothety of ratio 2 sending (ALM) to $(AXY) = \Omega$, so these two circles are tangent at A . Hence, RA is also tangent to Ω as desired.

Solution 2 (Mathys): We give an alternative way of proving the Claim, using spiral similarity.

Let P be the center of the spiral similarity sending XA to BC . Note that this spiral similarity sends the respective midpoints L and K to each other. Therefore PXL and PBK are similar triangles, and in particular

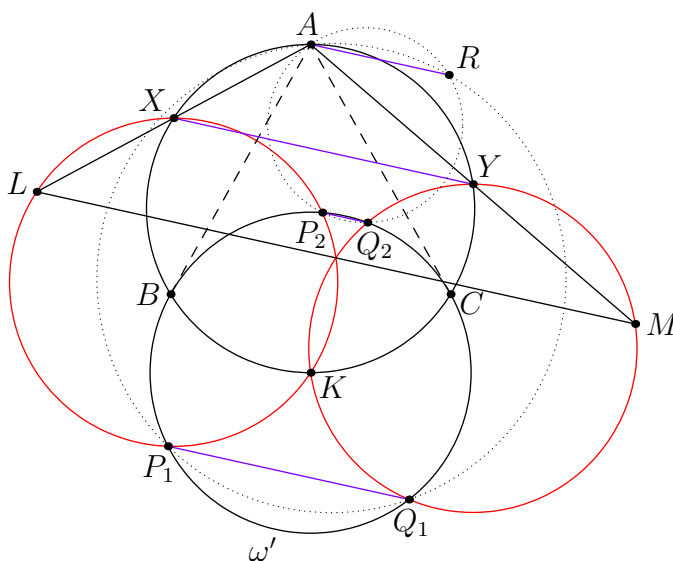
$$\angle PKX = \angle PKB = \angle PLX,$$

meaning that $XLKP$ is cyclic. Now, as $LX = LK$, we know that $\angle XPL = \angle LPK$ by WUM, and since $\angle XPL = \angle BPK$ by spiral similarity, we immediately have L, B, P collinear. Lastly, the spiral similarity between PXA and PBC implies that $XACP$ is cyclic, and that means that

$$\angle CPB = \angle APX = \angle ACX = \angle ACB.$$

By the tangent chord theorem, (BPC) is tangent to C , and so P must lie on ω . This proves $P = P_2$.

Solution 3 (Felix): We can invert the configuration at a circle centered at A with arbitrary radius. For the sake of conciseness, we will still call the image of points by their actual names (writing B instead of B^* for example).



Note that $AXBKCY$ is cyclic since $XBKCY$ collinear in the original configuration. Furthermore K is the antipode of A and $AB = AC$. Note that $AL = 2AX$ and $AM = 2AY$ since they were the midpoints before inversion.

Therefore, by perpendicularity and midpoints, KX is the perpendicular bisector of AL , and

hence (KXL) is the circle (ABC) mirrored at KX . Similarly, (KYM) is the circle (ABC) mirrored at KY . Note that ω gets mapped to the circle ω' around K with radius $KB = KC$. Since $\{P_1, P_2\} = (KXL) \cap \omega'$, P_2 is the reflection of B across KX and P_1 is the reflection of C across KX , similarly we get Q_1 and Q_2 . In particular, that means that $ALBP_2$ (and symmetric quadrilaterals) are cyclic, which is exactly the inverted counterpart of the claim in Solution 1.

We now show that $XY \parallel P_1Q_1$ and $XY \parallel P_2Q_2$. This can be done by angle chase in the following manner:

$$\begin{aligned}\angle Q_2YX &= \angle KYX - \angle KYQ_2 \\ &= \angle KAX - \angle CYK \\ &= \angle KAB + \angle BAX - \angle CYK \\ &= \angle BAX = \angle BYX\end{aligned}$$

implies that B, Q_2, Y are collinear, and similarly C, P_2, X are collinear, so we have

$$\angle CP_2Q_2 = \angle CBQ_2 = \angle CBY = \angle CXY$$

by cyclicity, which concludes (other, more lengthy angle chases are possible).

Note that $\omega' = (P_1Q_1Q_2P_2)$, ARP_1Q_1 cyclic and ARP_2Q_2 cyclic. Therefore AR , P_1Q_1 and P_2Q_2 concur by radical center. But the radical center is the point at infinity parallel to P_1Q_1 (since $P_1Q_1 \parallel P_2Q_2$), thus $AR \parallel P_1Q_1 \parallel XY$. Translating back to the original world, this tells us that AR is tangent to circle (AXY) in our original configuration.

Marking Scheme

The marking schemes are additive.

Solution 1:

- (A) **4P**: Part I (Collinearity of L, B, P_2 and symmetric triples)
 - (A1) **1P**: Claiming that L, B, P_2 are collinear.
 - (A2) **1P**: Noticing that L lies on the radical axis of (A) and ω (or equivalently, that $\text{Pow}_\omega(L) = LA^2$).
 - (A3) **1P**: Stating that $LX = LK$.
 - (A4) **1P**: Introducing P as $LB \cap \omega$ and proving that $XLKP$ is cyclic (or a similar approach).
- (B) **2P**: Part II (Power of a point considerations)
 - (B1) **1P**: Proving that R lies on LM .
 - (B2) **1P**: Proving that $RA^2 = RL \cdot RM$, or an equivalent statement.
- (C) **1P**: Conclusion

Solution 2:

- (A) **4P**: Part I (Collinearity of L, B, P_2 and symmetric triples)
 - (A1) **1P**: Claiming that L, B, P_2 are collinear.
 - (A2) **1P**: Introducing P as the center of spiral similarity between XA and BC (or an equivalent definition).
 - (A3) **1P**: Proving that P lies on Ω and LB .
 - (A4) **1P**: Proving that P lies on ω .
- (B) **2P**: Part II (Power of a point considerations): same as in Solution 1
- (C) **1P**: Conclusion

For an inverted solution, we take the same marking schemes (with the inverted counterparts or equivalents for each statement).

In all solutions: if no points from (B) are awarded, but the student claimed one of the following:

- R lies on LM **and/or**
- P_1Q_1LM and P_2Q_2LM are cyclic **and/or**
- AP_1Q_1 and AP_2Q_2 are tangent to Ω ,

they get **1P** for this part.



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Second Exam – Solutions

Duration: 4.5 hours

Difficulty: The problems are ordered by difficulty.

Points: Each problem is worth 7 points.

Bern

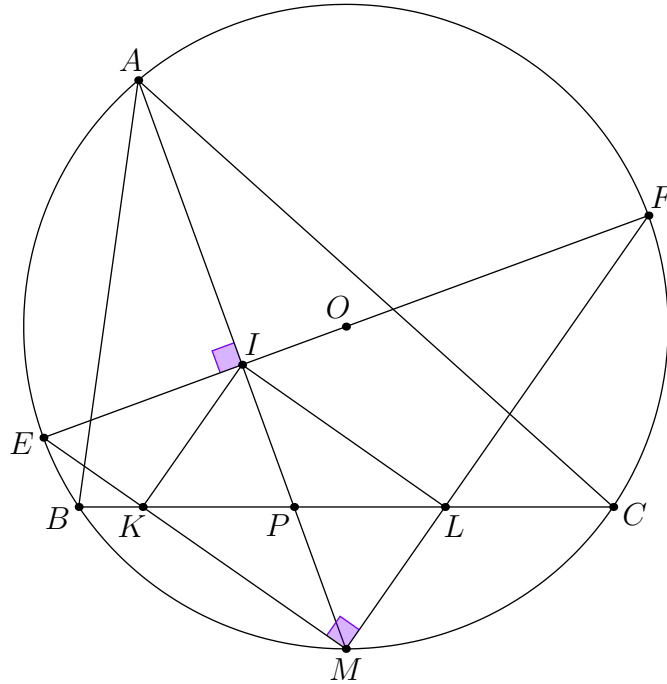
May 11, 2025

Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

4. Let ABC be a triangle with $AB < AC$. Denote its circumcircle by Ω , its circumcenter by O , and its incenter by I . Let M be the midpoint of the arc BC not containing A . Line OI meets Ω at E and F , and line BC meets ME and MF at K and L , respectively. Suppose that $IA = IM$. Prove that $IKML$ is a rectangle.

Solution



From the length condition, we have that OI is the perpendicular bisector of AM , in particular $\angle AIE = 90^\circ$. Moreover EF is a diameter of Ω , and $\angle LMK = \angle FME = 90^\circ$.

Consider the inversion with center M and radius MB . By the incenter/excenter lemma, it fixes B , I , and C , and it sends line BC onto Ω : in particular it exchanges both K and E , and L and F . Hence $\angle MKI = \angle MIE = 90^\circ$ and $\angle ILM = \angle FIM = 90^\circ$. This finishes the proof that $IKLM$ is a rectangle.

Solution 2: Let $P = AM \cap BC$. We first prove that $PK = PM = PL$ by angle chase. As EF is the perpendicular bisector of AM , we have

$$\angle PMK = \angle AME = \frac{1}{2} \cdot \angle ACM = \frac{\alpha}{4} + \frac{\gamma}{2},$$

and

$$\angle KPM = \angle BPM = \frac{\alpha}{2} + \beta,$$

so $\angle MKP = 180^\circ - \frac{\alpha}{4} - \frac{\gamma}{2} - \frac{\alpha}{2} - \beta = \frac{\alpha}{4} + \frac{\gamma}{2} = \angle PMK$ and $PK = PM$. Similarly, $PL = PM$.

Moreover, as $\triangle MPB \sim \triangle MBA$ and by the incenter/excenter lemma,

$$\frac{MP}{MI} = \frac{MP}{MB} = \frac{MB}{MA} = \frac{MI}{MA} = \frac{1}{2}.$$

Hence $PK = PM = PL = PI$ and $IKML$ is a rectangle.

Marking Scheme

Solution 1 (angles):

- (A) **1P:** Showing that $\angle FME = 90^\circ$.
- (B) **1P:** Showing that $AM \perp EF$.
- (C) **4P:** Showing that $\angle MKI = 90^\circ$ or/and that $\angle ILM = 90^\circ$.
 - (C1) **1P:** Considering the inversion centered at M with radius MB .
 - (C2) **1P:** Showing that $\triangle MKB \sim \triangle MBE$ or that $\triangle MLC \sim \triangle MCP$.
 - (C3) **1P:** Showing that $MB^2 = MK \cdot ME$ or that $MC^2 = ML \cdot MF$.
 - (C4) **2P:** Showing that $MI^2 = MK \cdot ME$ or that $MI^2 = ML \cdot MF$.
- (D) **1P:** Concluding

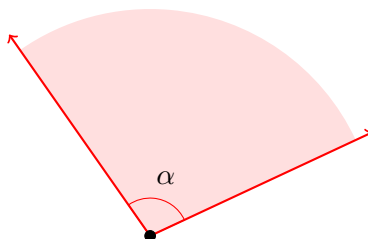
Solution 2 (distances):

- (A) **0P:** Showing that $\angle FME = 90^\circ$.
- (B) **1P:** Showing that $AM \perp EF$.
- (C) **3P:** Showing that $PK = PM$ or/and that $PL = PM$.
 - (C1) **1P:** Computing a useful expression for $\angle PMK$ or $\angle LMP$.
 - (C2) **1P:** Computing a useful expression for $\angle KPM$ or $\angle MPL$.
 - (C3) **1P:** Showing that $EKLF$ is cyclic.
 - (C4) **2P:** Showing that $\angle MKP = \angle PMK$ or that $\angle PLM = \angle LMP$.
- (D) **2P:** Showing that $PI = PK$, or $PI = PM$, or $PI = PL$.
 - (D1) **1P:** Considering the inversion centered at M with radius MB .
 - (D2) **1P:** Showing that $\triangle MPB \sim \triangle MBA$ or that $\triangle MPC \sim \triangle MCA$.
 - (D3) **1P:** Showing that MB is tangent to (PBA) .
 - (D4) **1P:** Showing that MC is tangent to (PCA) .
 - (D5) **1P:** Showing that EF is tangent to (IKL) .
- (E) **1P:** Concluding.

Remarks: The marking schemes are additive, but the subitems are non-additive. Each exercise should be graded with the marking scheme that awards the most points. For example, a student that only shows that $\angle FME = 90^\circ$ and $AM \perp EF$ is awarded 2 points. The point for concluding can only be awarded to a full solution (possibly with minor mistakes).

5. Let α be a real number satisfying $0 < \alpha < 180$. For Leo's birthday, Frieder has placed 2025 gnomes at arbitrary points inside his garden. No three gnomes are collinear and no two gnomes coincide. Each gnome has a field of view spanning α degrees (including the boundary). After Frieder places the gnomes down, Leo wants to rotate the gnomes such that, for each gnome, the number of other gnomes it sees is different.

Determine all values of α for which Leo can achieve this, regardless of how the gnomes are placed.



An example of the field of view of a gnome. It extends infinitely between the boundary rays.

Solution

Answer: Any α in the interval $[180 \cdot \frac{2023}{2025}, 180[$.

The proof that values less than $180 \cdot \frac{2023}{2025}$ do not work is common to all proofs: simply consider the construction where the vertices are arranged on the vertices of a regular 2025-gon. One gnome must see all the others (there are 2025 possible numbers of gnomes one can see and 2025 gnomes, so we must have a bijection) and this gnome's field of view must span at least the interior angle of a regular 2025-gon, or $(180 \cdot \frac{2023}{2025})^\circ$. Now we proceed to the proof that any larger angle is sufficient.

Proof 1: Number the gnomes $G_1, G_2, \dots, G_{2025}$. Denote by S_n the set of values for the number of other gnomes G_n can possibly see (i.e., if you rotate G_n , what different numbers of gnomes will it have in its field of view over the course of a full rotation?). Note that S_n is an interval (of integers) because as the values change (when gnomes drop in and out of the field of view of G_n) they change by at most 1 between two consecutive values (since two gnomes can't enter or exit the field of view at the same time, due to the non-collinearity condition) implying that S_n must contain every integer between its minimum and its maximum.

Now, we seek to prove the following lemma: for any set of k gnomes $G_{j_1}, G_{j_2}, \dots, G_{j_k}$, we have

$$\left| \bigcup_{i=1}^k S_{j_i} \right| \geq k.$$

If this is true (i.e. if any set of k gnomes can see at least k different values of other gnomes) then we are done by Hall's Marriage Theorem, as this implies the existence of a bijective mapping between gnomes and values.

Consider an arbitrary set of k gnomes. Looking at their convex hull, there is at least one gnome G_{j_n} on the boundary of this polygon whose corresponding angle is less than or equal to $(180 \cdot \frac{k-2}{k})^\circ$, which in turn is less than or equal to α , so we can rotate it to see all $k-1$ others. Rotate the other $k-1$ gnomes to face the same way as G_{j_n} . Now, there is at least one gnome G_{j_m} that sees none of the $k-1$ others (when gnomes face the same direction, they cannot both

see each other as $\alpha < 180$). Additionally, the field of view of G_{j_m} is a strict subset of G_{j_n} , so if G_{j_n} currently sees N_1 gnomes in total and G_{j_m} sees M_1 gnomes in total, we have

$$N_1 - M_1 \geq k - 1.$$

Now, if we rotate both G_{j_n} and G_{j_m} to face the opposite direction, we have that the field of view of G_{j_n} instead becomes a strict subset of G_{j_m} . So if they now see N_2 and M_2 gnomes respectively, we have

$$M_2 - N_2 \geq 1.$$

Now we have two cases. Firstly, if $M_2 > N_1$ and/or $N_2 < M_1$, then the cardinality of the set S_{j_m} and/or the set S_{j_n} respectively is at least k , and we are done. Otherwise, we have $M_1 \leq N_2 \leq M_2 \leq N_1$ and we must then have that (since both sets are intervals)

$$|S_{j_m} \cup S_{j_n}| \geq (N_1 - M_1) + 1 = k$$

and we are again done.

Marking Scheme

Full solutions are awarded 7 points, with up to 2 points removed at the discretion for the coordinators for flaws (e.g. the remark in B1 below). Partial solutions are awarded points according to the following markscheme.

- (A) **1P:** (Construction) Showing a field of view of at least $(180 \cdot \frac{2023}{2025})$ degrees is *necessary* for one gnome in a set of 2025 to see all the others because of the regular 2025-gon.
- (B) **4P:** (Setup) Showing that k gnomes see at least k different values.
 - (B1) **1P:** Showing that the set of values of the number of other gnomes a given gnome can see is an interval. Non-collinearity must not necessarily be mentioned (but if it is omitted from an otherwise perfect solution, 1 point will be deducted).
 - (B2) **1P:** Proving that any set of k gnomes, it is possible to orient two of them such that one sees at least $k - 1$ more gnomes (over all the gnomes, not just the set of k) than the other.
 - (B3) **1P:** Proving that for any two gnomes, the respective interval of values they can see have nonempty intersection. It is sufficient to point this out for just the two gnomes described in B2, if B2 was already obtained.
 - (B4) **1P:** Having all three of B1, B2 and B3 and concluding that k gnomes see at least k different values.
- (C) **2P:** (Finish) Stating that if the cardinality of the union over k gnomes of these sets is at least k , we can conclude via Hall's Marriage Theorem. Providing the precise name of the result is not necessary as long as the participant makes it clear they know such a result exists.

6. Let $a_1, a_2, a_3, \dots$ be an infinite sequence of real numbers satisfying, for all integers $n \geq 1$,

$$a_{\lfloor \frac{n}{1} \rfloor} \cdot a_{\lfloor \frac{n}{2} \rfloor} \cdot \dots \cdot a_{\lfloor \frac{n}{n} \rfloor} = 2^{n^2}.$$

Prove that $\frac{a_{n+1} - a_n}{n+1}$ is an integer for all $n \geq 1$.

Solution

First solution :

Let b_n the number of ordered pairs (x, y) of positive coprime integers smaller or equal to n .

We can double count the number of pairs (x, y) of positive integers smaller or equal to n according to their greatest common divisor : there is a bijection between the pairs (x, y) which have their greatest common divisors equal to d and the pairs (x', y') of coprime integers smaller or equal to $\lfloor \frac{n}{d} \rfloor$ by $x \rightarrow x' = \frac{x}{d}, y \rightarrow y' = \frac{y}{d}$. This proves that

$$n^2 = \sum_{d=1}^n b_{\lfloor \frac{n}{d} \rfloor}$$

and therefore $a_n = 2^{b_n}$ as a_n is uniquely determined.

In particular, $b_n - b_{n-1} = 2\varphi(n)$ because the coprime pairs which have exactly one component equal to n must have their other component coprime to n .

In that case, $a_n - a_{n-1} = (2^{b_n - b_{n-1}} - 1)2^{b_{n-1}} = (4^{\varphi(n)} - 1)a_{n-1}$. Write $n = 2^m k$ where $k \in \mathbb{N}$ is odd and $m \geq 0$. Then, by Euler's theorem, $4^{\varphi(n)} = (4^{\varphi(k)})^{\varphi(2^m)} \equiv 1^{\varphi(2^m)} \equiv 1 \pmod{k}$ and

$$\nu_2(a_{n-1}) = 2\varphi(1) + 2\varphi(2) + \dots + 2\varphi(n) - 1 \geq n \geq 2^m \geq m$$

since $\varphi(r) \geq 1$ for any $r \in \mathbb{N}$ so 2^m divides a_{n-1} and as k divides $4^{\varphi(n)} - 1$, $n = 2^m k$ divides $(4^{\varphi(n)} - 1)a_{n-1} = a_n - a_{n-1}$.

Second solution :

Clearly $a_n \neq 0$ for every $n \in \mathbb{N}$. Divide the equality for $n+1$ by the equality for n to get

$$a_{\lfloor \frac{n+1}{n+1} \rfloor} \prod_{k=1}^n \frac{a_{\lfloor \frac{n+1}{k} \rfloor}}{a_{\lfloor \frac{n}{k} \rfloor}} = 2^{2n+1}.$$

By plugging $n=1$ in the equation we get $a_1 = 2$. Furthermore, for any integers $1 \leq k \leq n$, by euclidean division we can write $n = ak + b$ where $a \in \mathbb{N}$ and $0 \leq b \leq k-1$. If $b \neq k-1$, then the euclidean division of $n+1$ by k is $ak + (b+1)$ where $0 \leq b+1 \leq k$ so we have $\lfloor \frac{n+1}{k} \rfloor = \lfloor \frac{n}{k} \rfloor = a$.

In that case $\frac{a_{\lfloor \frac{n+1}{k} \rfloor}}{a_{\lfloor \frac{n}{k} \rfloor}} = 1$.

If $b = k-1$, then the euclidean division of $n+1$ by k is $(a+1)k$ therefore $\lfloor \frac{n+1}{k} \rfloor = \lfloor \frac{n}{k} \rfloor + 1 = a+1$.

With this, we can simplify the equality we have got above and we get

$$\prod_{k=1, k|n+1}^n \frac{a_{\frac{n+1}{k}}}{a_{\frac{n+1}{k}-1}} = \prod_{d=2, d|n+1}^{n+1} \frac{a_d}{a_{d-1}} = 2^{2n} = 4^n.$$

When $n+1 =: p$ is prime, this simplifies to

$$\frac{a_p}{a_{p-1}} = 4^{p-1} = 4^{\varphi(p)}.$$

When $n + 1 =: p^k$ is the power of a prime, this simplifies to

$$\prod_{m=1}^k \frac{a_{p^m}}{a_{p^m-1}} = 4^{p^k-1}.$$

Using this, we can prove by induction on k that $\frac{a_{p^k}}{a_{p^k-1}} = 4^{\varphi(p^k)}$. This motivates us to claim that $\frac{a_n}{a_{n-1}} = 4^{\varphi(n)}$ for every $n > 1$. Since the sequence $(a_i)_{i=1}^\infty$ is uniquely defined, this means it suffices to check that the sequence

$$b_n = \frac{1}{2} 4^{\varphi(1)+\varphi(2)+\dots+\varphi(n)}$$

satisfies the equation for every $n \geq 1$ in order to show that $a_n = b_n$. To check that, it is equivalent to check that (b_i) satisfies the equation for $n = 1$ (which is easily checked) and that (b_i) satisfies the equation for $n + 1$ divided by the equation for n , that is

$$\prod_{d=2, d|n+1}^{n+1} \frac{b_d}{b_{d-1}} = 4^n.$$

But $\frac{b_d}{b_{d-1}} = 4^{\varphi(d)}$ so it suffices to check that

$$\sum_{d=1, d|n+1}^{n+1} \varphi(d) = n + 1$$

which is a well-known formula (you can prove by induction on k the case of prime powers of the form p^k and use multiplicity to conclude the general case), so $a_n = b_n$ for every $n \geq 1$.

In that case, $a_n - a_{n-1} = (4^{\varphi(n)} - 1)a_{n-1}$ where a_{n-1} is an integer. Write $n = 2^m k$ where $k \in \mathbb{N}$ is odd and $m \geq 0$. Then, by Euler's theorem, $4^{\varphi(n)} = (4^{\varphi(k)})^{\varphi(2^m)} \equiv 1^{\varphi(2^m)} \equiv 1 \pmod{k}$ and

$$\nu_2(a_{n-1}) = 2\varphi(1) + 2\varphi(2) + \dots + 2\varphi(n) - 1 \geq n \geq 2^m \geq m$$

since $\varphi(r) \geq 1$ for any $r \in \mathbb{N}$ so 2^m divides a_{n-1} and as k divides $4^{\varphi(n)} - 1$, $n = 2^m k$ divides $(4^{\varphi(n)} - 1)a_{n-1} = a_n - a_{n-1}$.

Marking Scheme

The marking scheme is non-additive.

- (A) **5P**: Proving that $a_n - a_{n-1} = (4^{\varphi(n)} - 1)a_{n-1}$ for every $n > 1$.
 - (A1) **1P**: Considering b_n the number of ordered pairs (x, y) of positive coprime integers smaller or equal to n .
 - (A2) **3P**: Showing that $n^2 = \sum_{d=1}^n b_{\lfloor \frac{n}{d} \rfloor}$.
 - (A3) **4P**: Showing that $a_n = 2^{b_n}$.
 - (A4) **1P**: Showing that $a_1 \prod_{k=1}^n \frac{a_{\lfloor \frac{n+1}{k} \rfloor}}{a_{\lfloor \frac{n}{k} \rfloor}} = 2^{2n+1}$.
 - (A5) **2P**: Showing that $\prod_{d=2, d|n+1}^{n+1} \frac{a_d}{a_{d-1}} = 4^n$ or $\prod_{k=1, k|n+1}^n \frac{a_{\frac{n+1}{k}}}{a_{\frac{n+1}{k}-1}} = 4^n$.
 - (A6) **3P**: Showing that $\frac{a_n}{a_{n-1}} = 4^{\varphi(n)}$ for an infinite family of positive integers $n > 1$.
 - (A7) **3P**: Claiming that $a_n = \frac{1}{2}4^{\varphi(1)+\varphi(2)+\dots+\varphi(n)}$ or that $\frac{a_n}{a_{n-1}} = 4^{\varphi(n)}$ for every $n > 1$.
- (B) **2P**: Finish.
 - (B1) **1P**: If we write $n = 2^m k$ where k is odd, showing that $2^m | a_n - a_{n-1}$.
 - (B2) **1P**: If we write $n = 2^m k$ where k is odd, showing that $k | a_n - a_{n-1}$.



IMO Selection 2025

Third Exam – Solutions

Duration: 4.5 hours

Difficulty: The problems are ordered by difficulty.

Points: Each problem is worth 7 points.

Bern

May 24, 2025

Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

7. Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(y) - yf(x)) = f(xy) - xy$$

holds for all $x, y \in \mathbb{R}$.

Solution

The only functions satisfying the equation are the identity function $f(x) = x$ and the absolute value function $f(x) = |x|$.

Assume that f satisfies the given equation. Plugging in $y = 0$ gives us

$$f(xf(0)) = f(0). \quad (1)$$

We either have that f is constant or that $f(0) = 0$. Clearly no constant function works, so we must have $f(0) = 0$. Plugging in $x = y$ gives us

$$f(0) = f(x^2) - x^2 \rightarrow f(z) = z \text{ for } z \geq 0. \quad (2)$$

Assume that f isn't the identity function, i.e. there is a $a \in \mathbb{R}$ with $f(a) \neq a$. Let $\lambda = \left| \frac{z}{f(a)-a} \right| \geq 0$. Plugging in $(x, y) = (\lambda, a)$ and $(x, y) = (a, \lambda)$ we have

$$f\left(\left|\frac{z}{f(a)-a}\right| \cdot (f(a) - a)\right) = f(\lambda a) - \lambda a = f\left(\left|\frac{z}{f(a)-a}\right| \cdot (a - f(a))\right) \Rightarrow f(z) = f(-z), \quad (3)$$

which readily implies that $f(x) = f(-x) = |x|$ must hold for all $x \in \mathbb{R}$.

All that is left to do now, is to check that the two functions $f(x) = x$ and $f(x) = |x|$ indeed work:

For $f(x) = |x|$, we distinguish between the following four cases:

$$\begin{aligned} x, y \geq 0 & \quad |xy - yx| = 0 = |xy| - xy \\ x, y < 0 & \quad |-xy + xy| = 0 = |xy| - xy \\ x \geq 0 > y & \quad |-xy - xy| = -2xy = |xy| - xy \\ x < 0 \leq y & \quad |xy + xy| = -2xy = |xy| - xy. \end{aligned}$$

That the identity function $f(x) = x$ works follows directly from the calculations of the first case.

Marking Scheme

- (A) **1P:** Claim the correct solution set and show that $f(x) = x$ and $f(x) = |x|$ indeed work.
- (B) **1P:** Show that $f(0) = 0$.
- (C) **1P:** Show that $f(x^2) = x^2$ for all x
- (D) **3P:** Correctly show that if f isn't the identity then $f(x) = f(-x)$ for all x .
 - (D1) **1P:** Showing injectivity at 0
 - (D2) **1P:** Showing linearity in the negatives

8. Let a and b be integers with $a > b \geq 5$. Prove that there exist a positive integer k and positive integers $c_1, c_2, \dots, c_k$ with $c_1 = a$ and $c_k = b$, such that $c_i^2 + c_{i+1}^2$ is a perfect square for all $1 \leq i < k$.

Solution

First solution : We work in the graph G whose vertices are the integers ≥ 5 and there's an edge between x and y iff $x^2 + y^2$ is a perfect square. It suffices to prove that G is connected.

We first show that in G , every integer x is connected by a path to some power of two at least x . Write x as $2^\alpha m$ for $m > 1$ odd, and let $p > 2$ be the largest prime dividing m . Now define,

$$x' = 2^\alpha \cdot \frac{m}{p} \cdot 2 \cdot \frac{p-1}{2} \cdot \frac{p+1}{2} = 2^{\alpha-1} \frac{m(p^2-1)}{p}$$

and observe that $x' > x$ and

$$x^2 + (x')^2 = 4^\alpha m^2 + 4^{\alpha-1} \frac{m^2(p^2-1)^2}{p^2} = \left(\frac{2^\alpha m(p^2+1)}{p} \right)^2$$

is a perfect square. Thus, there is an edge between x and x' . If we now write $x' = 2^{\alpha'} m'$ with m' odd, notice that $v_p(m') = v_p(m) - 1$. And that m' has no prime divisor larger than p since $m' \mid p^2 - 1 = (p-1)(p+1)$. Thus, repeatedly replacing x by x' $v_p(m)$ many times, we end up with an integer x^* that is connected to x in G and whose maximal prime divisor is strictly smaller than p . Now repeatedly replacing x by x^* , we arrive at an integer y that is still connected to x in G , and whose largest prime divisor is two. Thus, x is connected to a power of two, and since $x' > x$ for every x , this power of two is at least x .

It remains to show that all powers of two beyond 4 are connected. The sequence

$$8 - 15 - 20 - 21 - 72 - 96 - 40 - 30 - 16$$

shows that 8 is connected to 16 in G . Scaling this sequence by 2^k for any positive integer k , we find that any two consecutive powers beyond 8 are connected, as desired.

Second solution : First note that if a, b are connected and c, d are connected in G , then ac is connected to bc which is connected to bd .

We can check by hand that 5, 6, 7, 8, 10, 30 are connected.

We prove by induction on $n \geq 8$ that the integers 5, 6, 7, $\dots$, n are connected in G .

Suppose the claim is true for $n-1$, then if n is even, then $n = 2 \cdot \frac{n}{2}$ is connected to $2 \cdot 5$ as $\frac{n}{2}$ is connected to 5

If $n = 2k-1$ is odd, then it is connected to $2k(k-1) = k(2k-2)$ which is connected to $5 \cdot 6 = 30$ which is connected to 5.

Marking Scheme

First Solution:

- (A) **4P:** Proving that every integer is connected to a power of 2
 - (A1) **2P:** Proving that an odd prime p is connected to a product of smaller primes, or showing that x is connected to x' where x' is defined as in the solution.
- (B) **2P:** Proving that all the powers of 2 are connected
 - (B1) **1P:** Connecting every powers of 2 except finitely many.
 - (B2) **1P:** Showing that, if 8 and 16 are connected, then we are done.

Second Solution:

- (A) **1P:** Proving that if a, b are connected to 5, then their product is also connected to 5.
- (B) **4P:** Proving the induction step
 - (B1) **2P:** Proving the induction step for n odd
 - (B2) **2P:** Proving the induction step for n even

9. Leo's garden has now been passed on to his niece Batlpász, who is a bit disappointed by her inheritance at first. However, one day, she gets a letter from Frieder telling her that there is buried treasure under one of the 2025 gnomes in the garden, but she doesn't know which one.

Some pairs of gnomes are directly connected by a dirt path, and at most three dirt paths meet at each gnome. Bãtlpász can get from any gnome to any other gnome by walking along some of the dirt paths. However, removing any single dirt path would split the gnomes into exactly two separate connected groups, with no way to get from one group to the other.

In the middle of each dirt path is a fairy that can tell Batlpász which direction along the path she has to go to get to the gnome with the treasure. Every morning for a fortnight, Batlpász may ask a fairy for help. After the fourteenth question, if Batlpász has not found the location of the treasure, it will vanish. Can Batlpász always find the treasure, no matter how the dirt paths are arranged?

Solution

Answer: No 14 questions is not enough.

We will reformulate our question in terms of graph theory, where each gnome represents a vertex and a dirt path connecting two gnomes is an edge. Since there exists exactly one unique path between every pair of vertices our graph is a connected tree with the degree of every vertex being at most three. Below we will show a construction of a possible tree and show Bea might be required to ask at least 15 questions.

Construction: Let T_0 be a tree rooted binary tree of depth 1 and T_1 be a binary tree with depth 2. For $n \geq 2$, let T_n be a tree that is rooted at a vertex with two children, where one of the children is the root of a copy of T_{n-1} and the other of T_{n-2} .

The order of $|T_0| = 3$ and $|T_1| = 7$. $|T_n|$ can be defined recursively by the formula $|T_n| = |T_{n-1}| + |T_{n-2}| + 1$ for all $n \geq 2$. We will consider the tree T_{12} . In T_{12} there is no vertex with a degree greater than 3 and a quick calculation shows that it contains no more than 2025 vertices.

i	$ T_i $
0	3
1	7
2	11
3	19
4	31
5	51
6	83
7	135
8	219
9	355
10	575
11	931
12	1507

We will first show that one needs to ask four times to figure out the vertex with the treasure in T_1 . Since T_1 is symmetric we can assume that in the worst case the fairy will say that the treasure is towards the root of the tree. By taking the remaining tree of vertices where the treasure might be located we will have at least a tree with a root and three children. Here at

least three more questions are required if the fairy's always tell the direction towards the new root. Therefore four questions are required to figure out the location of the treasure in T_1 .

We will now show that if we have a tree which contains T_{i+1} as a subgraph then after one question it is always possible that we will have a tree T_i of possible locations of the treasure, for $i \geq 1$. Consider the root of the tree T_{i+1} it will have two children which are the roots of the T_i and T_{i-1} . Call the root of T_i , x . Notice that the degree of x is three and the three subtrees contain T_{i-1} , T_{i-1} , and T_{i-2} as subgraphs. This holds for $i \geq 2$ by our recursive construction. For $i = 1$ our three subtrees of vertex x are T_0 , T_0 , T_0 as T_1 contains two copies of T_0 as its children of its root (also x).

Notice after that after every question if the fairy points into the direction away from x in a tree T_{i+1} , then we will always have T_{i-1} and T_{i-2} (or T_{i-1} and T_{i-1}) of possible locations of the treasure which can be merged into a tree T_i rooted at x . Therefore by repeating this process we can see that after 11 questions on a tree T_{12} it is always possible to have a subgraph T_1 which consists of possible locations of the treasure. Since it takes 4 questions to guarantee to locate the treasure in T_1 it is possible that $11 + 4 = 15$ questions are required to locate the treasure in T_{12} .

Since the problem statement requires our tree to have more vertices than T_{12} , we can add all the extra vertices as a long path from any leaf in T_{12} . In general replacing an edge in our construction with two edges joined by a vertex will never decrease the required number of questions, as we can assume the treasure to not lie on joining vertex and both edges will now behave identically.

Remark: There exist other constructions which also work, similarly most of them require a recursive construction of the form $T_{i+1} = T_i + T_{i-1}$. For example taking T_0 a single vertex and T_1 a binary tree from three vertices produces a tree which requires 1973 vertices to create a tree that requires 15 questions.

Marking Scheme

A full solution is awarded 7 points. Up to 2 points can be removed for minor flaws. Incomplete solutions are graded with the following marking scheme.

- (A) **0P:** Claim that the answer is no.
- (B) **1P:** Provide a recursive description for a tree T_i for which after a question the tree of possible locations of where the treasure can be, is at least the size of T_{i-1} and prove that is the case.
- (C) **1P:** Prove that 14 questions could be needed for a construction (it is not necessary for said construction to require 15 questions).
- (D) **4P:** Provide a construction for which 14 questions is not enough to always find the treasure and claim this to be the case.
 - (D1) **2P:** Provide a recursive description for a tree that is created recursively by $T_i \approx T_{i-1} + T_{i-2}$

Partial score = ((A) + (B) + (C) + MAX(D, D1))

If a contestant provides and proves a construction for a tree that uses less than 2025 vertices and does not argue that adding extra vertices does not change the answer a point can be removed if they were to receive 7 points.

Minor arithmetic mistakes are not penalized.



IMO Selection 2025

Fourth Exam – Solutions

Duration: 4.5 hours

Difficulty: The problems are ordered by difficulty.

Points: Each problem is worth 7 points.

Bern

May 25, 2025

Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

10. Determine all $n \in \mathbb{N}$ with at least four positive divisors satisfying the following property:
For any distinct positive divisors $a, b \notin \{1, n\}$, we have $\gcd(a^b + 1, n) > 1$.

Solution

Answer: Only $n = 6$ works.

Proof 1: The most natural substitution to make is an a with as many prime divisors in common with n as possible, as none of these can divide $a^b + 1$, which is very restrictive (indeed, if there is a choice of a which shares all its prime divisors with n we have an instant contradiction, so n must be squarefree).

An appealing choice of a is therefore n/p_i for some prime divisor p_i of n . It follows that, for any divisor $b \neq n/p_i$ of n , we have

$$\begin{aligned} p_i &| (n/p_i)^b + 1 \\ \implies (n/p_i)^b &\equiv -1 \pmod{p_i} \end{aligned}$$

As n has at least 4 divisors, we can always take $b = p_i$, which by Euler-Fermat immediately implies $n/p_i \equiv -1 \pmod{p_i}$. So to not obtain an immediate contradiction, we must not be able to choose an even b and a p_i such that $p_i > 2$ and $n/p_i \neq b$. It follows that n is either odd or of the form $2p$, as otherwise $b = 2$ and a equal to n/p_i for an odd prime p_i would provide a contradiction.

Let us deal with the latter case and indeed eliminate the more general case where n is equal to the product of two primes $p_1 p_2$, with $p_1 < p_2$. Taking $a = p_1, b = p_2$ implies $p_2 | (p_1)^{p_2} + 1$ but the right expression is equal to $(p_1 + 1) \pmod{p_2}$, which can only happen if $p_2 = p_1 + 1$. So the only possibility is $n = 2 \cdot 3$, which we can easily check works.

We now move forward to the case where n is odd and has at least three prime divisors. Another appealing choice of a (since n has at least three prime divisors) is $n/p_i p_j$ with $p_i < p_j$. We must either have

$$(n/p_i p_j)^b \equiv -1 \pmod{p_i}$$

or

$$(n/p_i p_j)^b \equiv -1 \pmod{p_j}$$

for any divisor $b \neq n/p_i p_j$ of n (possibly both). Take $b = p_i$. In the first case, we would then have

$$\begin{aligned} (n/p_i)^{p_i} (1/p_j)^{p_i} &\equiv -1 \pmod{p_i} \\ \implies (1/p_j)^{p_i} &\equiv 1 \pmod{p_i} \\ \implies (1/p_j) &\equiv 1 \pmod{p_i} \\ \implies p_j &\equiv 1 \pmod{p_i} \end{aligned}$$

where the first implication uses the previously shown fact that $n/p_i \equiv -1 \pmod{p_i}$. In the second case we would instead have

$$\begin{aligned} (n/p_j)^{p_i} (1/p_i)^{p_i} &\equiv -1 \pmod{p_j} \\ \implies (1/p_i)^{p_i} &\equiv 1 \pmod{p_j} \end{aligned}$$

where, again, the first implication uses the previously shown fact that $n/p_i \equiv -1 \pmod{p_i}$. So in particular, the order of $(1/p_i)$ divides p_i and also $p_j - 1$ (Euler-Fermat). If the order is 1 then $(1/p_i) \equiv 1 \pmod{p_j}$, so $p_i \equiv 1 \pmod{p_j}$ which is impossible. Therefore the order is p_i and we again get $p_j \equiv 1 \pmod{p_i}$.

This is true for all pairs of primes. In particular, the largest prime p_m is 1 modulo any smaller prime dividing n . So $(p_m)^b + 1$ is 1 mod p_m and 2 modulo any smaller prime, contradiction.

Proof 2 (David): As above, it's easy to conclude that n is not squarefree (although now we will only need that it is not a prime power). Take $a = p_1$, the smallest prime divisor, and $b = p_m$, the largest prime divisor. Then some prime $p_k \neq p_1$ that divides n must also divide $(p_1)^{p_m} + 1$. In particular, $(p_1)^{2p_m} \equiv 1 \pmod{p_k}$. As the order of p_1 modulo p_k must divide $2p_m$, either it is 1 (absurd) or 2. This is as p_m and $2p_m$ are both bigger than $p_k - 1$ and cannot be the order. So $p_k \mid (p_1)^2 - 1 = (p_1 + 1)(p_1 - 1)$ and the only possibility is $p_k = 3, p_1 = 2$.

Letting $n = 6\ell$, we then have $(2\ell)^2 + 1$ is coprime with n so if $\ell \neq 1$ we have a contradiction by taking $a = 2\ell, b = 2$.

Marking Scheme

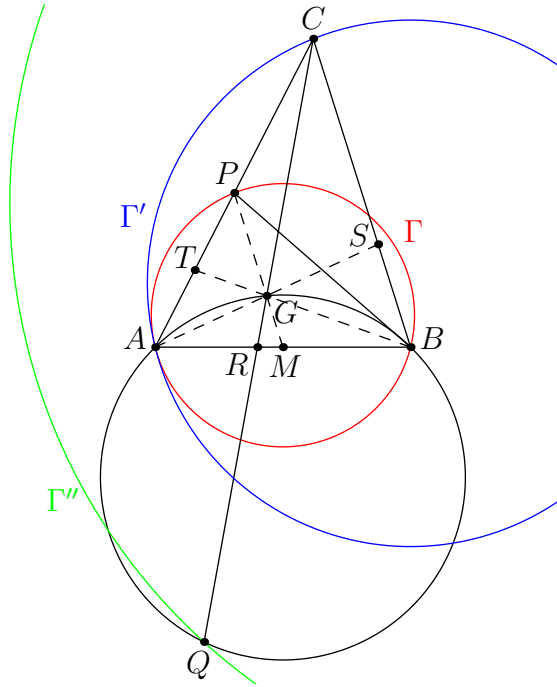
The essence of the problem is that it is easy to reduce to n being odd, squarefree and having at least 3 prime divisors. Partial solutions will therefore be rewarded by the following markscheme. In what follows, the prime divisors of n are denoted by $p_1 < p_2 < \dots < p_m$ with p_i, p_j being arbitrarily chosen divisors and $p_k \neq p_1$ being a prime divisor of $\gcd(n, (p_1)^{p_m} + 1)$.

- (A) **2P:** Completing the problem for the case when n is even.
- (B) **5P:** n is odd, >2 prime divisors, approach 1.
 - (B1) **1P:** Note $p_i \mid (n/p_i)^b + 1$ is forced.
 - (B2) **+1P:** Prove $(n/p_i) \equiv -1 \pmod{p_i}$.
 - (B3) **+2P:** Prove $p_j \equiv 1 \pmod{p_i}$ for $i < j$.
 - (B4) **+1P:** Conclude $((p_m)^b, n) = 1$.
- (C) **5P:** n is odd, >1 prime divisor, approach 2.
 - (C1) **1P:** Note the existence of $p_k \mid (p_1)^{p_m} + 1$.
 - (C2) **+2P:** Prove the order of $p_1 \pmod{p_k}$ is 2.
 - (C3) **+2P:** Conclude $p_1 = 2, p_k = 3$.
- (D) **1P:** Bonus partials
 - (D1) **1P:** Completing the problem for the case when n is not squarefree.
 - (D2) **1P:** Completing the problem for the case when n has two prime divisors exactly.

D1 and D2 are not additive with each other or any other points. Points within B and C are additive, and the contestant is awarded either just A, or the maximum of their points from either B or C with one additional point if they also did A.

11. Let Γ be a fixed circle with two fixed points A, B on it. For a variable point $P \notin \{A, B\}$ on Γ , let G be the center of gravity of the triangle ABP . The parallel to line GP through B intersects AP at point C , and the line CG intersects the circumcircle of the triangle ABG a second time at point Q . Prove that as P varies, the point Q lies on a fixed circle.

Solution



Let M be the midpoint of AB . First, we note that P is the midpoint of AC , as by Thales's theorem

$$\frac{AP}{AC} = \frac{AM}{MB} = \frac{1}{2}.$$

Moreover, if $R = AB \cap CG$, $S = BC \cap AG$ and $T = CA \cap BG$, then we can use Ceva's theorem to compute the signed ratio

$$\frac{AR}{RB} = \frac{AT}{TC} \cdot \frac{CS}{SB} = \frac{AT}{TC} \cdot \frac{PG}{GM} = \frac{1}{3} \cdot \frac{2}{1} = \frac{2}{3},$$

and in particular R is a fixed point.

By Menelaus on triangle RCB , we can also compute the signed ratio

$$\frac{RG}{GC} = -\frac{RA}{AB} \cdot \frac{BS}{SC} = \frac{2}{5} \cdot \frac{1}{2} = \frac{1}{5},$$

so that $RG = \frac{1}{6}RC$. Finally we come to point Q , where we can use the power of the point R with respect to the circle $(ABGQ)$ to find that

$$RQ = -\frac{RA \cdot RB}{RG} = -\frac{\frac{2}{5}AB \cdot \frac{3}{5}AB}{\frac{1}{6}RC} = -\frac{36AB^2}{25} \cdot \frac{1}{RC}.$$

To conclude, we start by observing that C is the image of P by the homothety centered at A with ratio 2, which sends Γ to another fixed circle Γ' . By the previous computation, Q is the image of C by the composition of an inversion of radius $\frac{6}{5}AB$ centered at R with a reflection around point R . This new transformation sends circle Γ' onto another fixed circle Γ'' because $R \notin \Gamma'$ (as $R \in]AB[$) and this finishes the proof.

Marking Scheme

The marking scheme is non-additive.

- (A) **1P:** Proving that P is the midpoint of AC and/or that C lies on the fixed circle Γ' .
- (B) **3P:** Proving that $R = AB \cap CG$ is fixed.
- (C) **4P:** Proving that C lies on the fixed circle Γ' and that $R = AB \cap CG$ is fixed.
- (D) **4P:** Proving that $R = AB \cap CG$ is fixed, and that RQ is a fixed multiple of $\frac{1}{RC}$.
- (E) **5P:** Proving that C lies on the fixed circle Γ' , that $R = AB \cap CG$ is fixed, and that RQ is a fixed multiple of $\frac{1}{RC}$.

The following deductions are worth -1P and are additive :

- No mention of $R \notin \Gamma'$ in the proof that Γ' is sent onto another circle Γ'' .
- No mention of composition with a point reflection (inversion instead of negative inversion).

12. Find all polynomials $P \in \mathbb{Z}[x]$ for which there exists a non-constant polynomial $Q \in \mathbb{Z}[x]$ satisfying the following:
For all integers a, b we have $P(b-a) \mid Q(b) - Q(a)$.

Solution

Answer: Polynomials $P(x) = c$ and $P(x) = cx$, with any $c \in \mathbb{Z}, c \neq 0$.

Clearly, the polynomials $P(x) = c, cx$ work for any $c \in \mathbb{Z}, c \neq 0$, by taking $Q(y) = cy$. Now, let P be such a polynomial, for which Q works. Clearly, $P \not\equiv 0$ because else it would imply that Q is constant (for every $a, b \in \mathbb{Z}$, $Q(a) = Q(b)$). We now give three ways of proving that P has to be in the above form.

Solution 1 (Mathys): We write $P(x) = x^k \cdot R(x)$, where $R(0) \neq 0$. Obviously, Q also works for R . For now, we suppose that R is nonconstant. Let $c = R(0) \neq 0$. We can write $R(x) = c + x \cdot S(x)$. We claim that for all integers a and infinitely many integers n , $R(n) \mid Q(a+c) - Q(a)$. Indeed, by Arnaud, $Q(a) \equiv Q(a+R(n)) \pmod{R(n)}$, and we also know that $Q(b+n) \equiv Q(b) \pmod{R(n)}$ by the condition of the problem. Therefore, by applying this result iteratively, $Q(b+kn) \equiv Q(b) \pmod{R(n)}$ for all $k \in \mathbb{Z}$ and thus

$$Q(a) \equiv Q(a+c+n \cdot S(n)) \equiv Q(a+c) \pmod{R(n)}.$$

But, assuming R is not constant, $R(n)$ takes arbitrarily large values and so $Q(a+c) = Q(a)$ for all $a \in \mathbb{Z}$. In other words, Q is periodic and it is a polynomial: it must be constant, which is a contradiction.

Now, let's focus on the x^k part to show that $k \leq 1$. To do this, it suffices to show that if $n^2 \mid Q(a+n) - Q(a)$ for all $a, n \in \mathbb{Z}$, then Q is constant. Notice that the above divisibility implies that, for a fixed $a \in \mathbb{Z}$, $n^2 \mid Q(a+n) - Q(a)$ as polynomials (because n^2 is monic). But, if we expand Q as $Q(x) = a_m x^m + a_{m-1} x^{m-1} + \dots + a_1 x + a_0$, we get that the coefficient of the degree 1 monomial is

$$a_m m a^{m-1} + a_{m-1}(m-1)a^{m-2} + \dots + a_1.$$

This is a polynomial in a , and it must vanish everywhere $\rightarrow$ therefore all the coefficients are zero, that is $a_i = 0$ for all $i \geq 1$, or in other words Q is constant. Contradiction, we're done.

Solution 2 (David): Assume $d = \deg P \geq 1$ (we know that all nonzero constant polynomials work), set $b = x + a$ and consider the quantity $\frac{Q(x+a)-Q(a)}{P(x)}$, which is an integer for infinitely many values of x (we ignore about the finitely many values of x with $P(x) = 0$). Considering a as fixed for now, both numerator and denominator are integer polynomials. By polynomial division, we can write

$$\frac{Q(x+a)-Q(a)}{P(x)} = S_a(x) + \frac{R(x)}{P(x)}$$

for some polynomials S_a and R with $\deg(R) < d$. We want to show that

$$S_a(x) + \frac{R(x)}{P(x)}$$

being an integer for infinitely many x implies that $R \equiv 0$.

Note that S_a and R need not have integer coefficients, but they have rational coefficients. Therefore, without loss of generality we can scale them by the lcm of the denominators appearing in the coefficients, and assume that they have integer coefficients, still wanting $S_a(x) + \frac{R(x)}{P(x)}$ to be an integer. Letting $x \rightarrow \infty$, we see that R must be the zero-polynomial, otherwise the LHS

wouldn't be an integer for sufficiently large x (indeed we have $|R(x)| < |P(x)|$ by comparing the degrees). So now, we have the important equation of polynomials:

$$\frac{Q(x+a) - Q(a)}{P(x)} = S_a(x), \text{ or in other words } Q(x+a) - Q(a) = P(x) \cdot S_a(x).$$

This equality must also hold for $x \in \mathbb{C}$ (the two sides are equal as polynomials), so if z is any root of P , we must have $Q(a+z) - Q(a) = 0$, and this is true for all $a \in \mathbb{Z}$. If $z \neq 0$, this would imply that Q is constant, which is not allowed! We conclude that $P(x) = cx^d$ for some $c \in \mathbb{Z}, c \neq 0$.

For the second (cool) part, consider some $a \in \mathbb{Z}$ where the derivative $Q'(a)$ does not vanish. We have

$$S_a(0) = \lim_{x \rightarrow 0} S_a(x) = \lim_{x \rightarrow 0} \frac{Q(x+a) - Q(a)}{P(x)} = \lim_{x \rightarrow 0} \frac{\frac{1}{x} \cdot (Q(x+a) - Q(a))}{c \cdot x^{d-1}}.$$

By assumption, the limit of the numerator exists and is not zero. Hence, $\lim_{x \rightarrow 0} c \cdot x^{d-1} \neq 0$, which implies $d = 1$. We easily check that all of those polynomials are indeed solutions.

Solution 3 (Mathys): With the same notation as in Solution 1, we can show that R is constant as follows:

By Schur's Theorem there exist infinitely many $p > |R(0)|$ such that $p \mid R(x_p)$ for some $x_p \in \mathbb{Z}$. That forces $p \nmid x_p$: indeed, if p divided x_p we would have by Arnaud that $p \mid R(0)$. Thus, we have

$$p \mid R(x_p) \mid Q(a+x_p) - Q(a),$$

for all $a \in \mathbb{Z}$, and by repeating that we obtain

$$Q(a) \equiv Q(a+x_p) \equiv Q(a+2x_p) \equiv \dots \equiv Q(a+(p-1)x_p) \pmod{p}.$$

But as $p \nmid x_p$, $a + ix_p$ runs over all p classes $\pmod{p}$ and we get that Q is constant modulo p . In other words, $p \mid Q - Q(0)$, for all such p that we defined above. This readily implies that fixing $n \in \mathbb{Z}$, infinitely many primes divide $Q(n) - Q(0)$, meaning that Q is constant, and this is a contradiction. Therefore R must be constant.

Marking Scheme

The marking scheme is non-additive.

- (A) **0P:** Proving that $P(x) = c$ or $P(x) = cx$ for nonzero $c \in \mathbb{Z}$ are solutions.
- (B) **4P:** Proving that P is a monomial.
 - (B1) **1P:** Writing P as $x^k \cdot R(x)$, with $R(0) \neq 0$.
 - (B2) **2P:** Proving that $R(n) \mid Q(a + R(0)) - Q(a)$ for all integers n .
 - (B3) **2P:** Proving that $P(x) \mid Q(x + a) - Q(a)$ as rational polynomials.
 - (B4) **2P:** Finding some $p \nmid x_p$ such that $p \mid Q(a + x_p) - Q(a)$ (assuming P is not a monomial).
 - (B5) **3P:** Proving that, if P is not a monomial, Q is periodic.
- (C) **3P:** Proving that x^2 doesn't divide P .
 - (C1) **1P:** Stating $n^2 \mid Q(a + n) - Q(a)$ has to hold as integer polynomials, if x^2 divides P .
 - (C2) **2P:** Proving that the linear coefficient of $Q(x + a)$ cannot be zero for all integers a .